

# TrES-5b

R-1280

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_211126 · created 2026-07-31T14:50:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459544.6299 +/- 0.0034 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.159 +/- 0.052                 |
| Transit depth (Rp/Rs)^2                           | 2.53 +/- 1.65 [%]               |
| Orbital Inclination (inc)                         | 88.94 +/- 2.74                  |
| Airmass coefficient 1 (a1)                        | 1.002 +/- 0.053                 |
| Airmass coefficient 2 (a2)                        | -0.002 +/- 0.035                |
| Scatter in the residuals of the lightcurve fit is | 5.32 %                          |
| Best Comparison Star                              | #1 - [471, 32]                  |
| Optimal Aperture                                  | 2.94                            |
| Optimal Annulus                                   | 6.98                            |
| Transit Duration (day)                            | 0.0855 +/- 0.0071               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.159 ± 0.052 vs 0.142 (deviation 0.33σ) |
| Duration fit vs published | 0.0855 d vs 0.0758 d (deviation 1.36σ)   |
| Mid-time O–C              | 0.5 min                                  |
| Depth SNR                 | 3.1                                      |
| Residual scatter          | 5.32 %                                   |

## Light curve

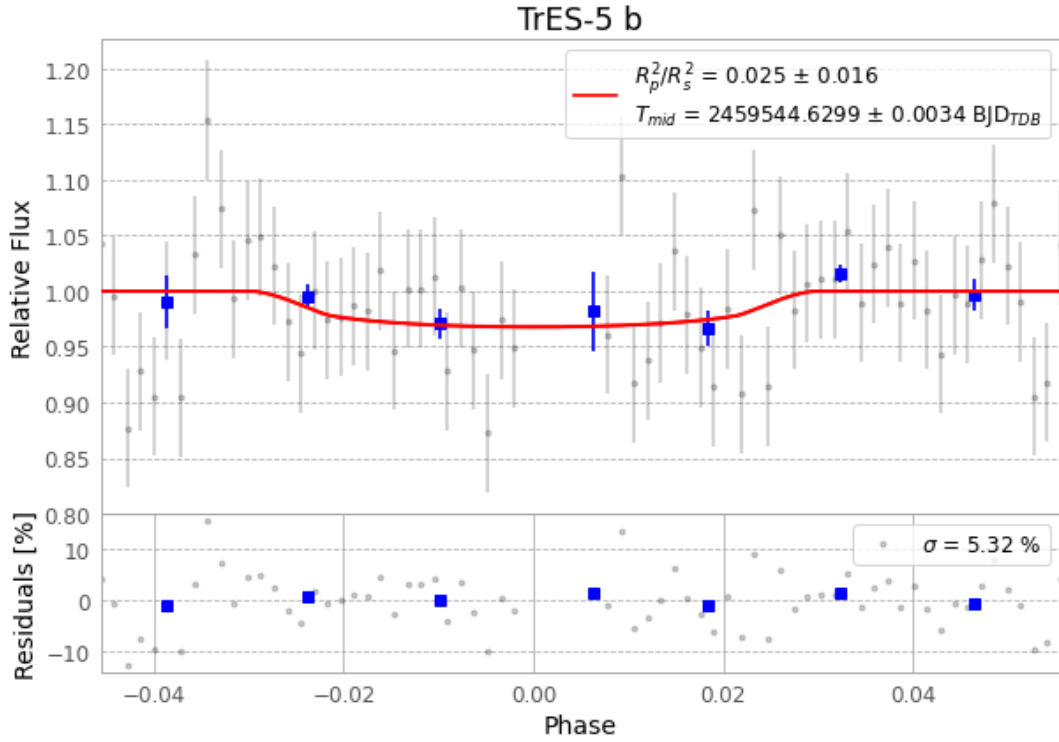


Figure 1 — FinalLightCurve\_TrES-5 b\_2021-11-25.png (SHA-256 11a0a80f6f4ab69b...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 275], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 fba67dda834c05f4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

75 FITS frames (49 MB), TRES-5211126012112.FITS → TRES-5211126050312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-211126.log (a739063c815b...), FinalLightCurve\_TrES-5 b\_2021-11-25.png (11a0a80f6f4a...), FinalLightCurve\_TrES-5 b\_2021-11-25.csv (a3373d3cac59...)

## Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:50Z.